

DS lab 10.c - Code::Blocks 25.03

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```
1  #include <stdio.h>
2  #include <stdlib.h>
3  struct Node
4  {
5      int data;
6      struct Node *left;
7      struct Node *right;
8  };
9  struct Node* createNode(int value)
10 {
11     struct Node *newNode;
12     newNode = (struct Node*)malloc(sizeof(struct Node));
13     newNode->data = value;
14     newNode->left = NULL;
15     newNode->right = NULL;
16     return newNode;
17 }
18 void preorder(struct Node *root)
19 {
20     if(root != NULL)
21     {
22         printf("%d ", root->data);
23         preorder(root->left);
24         preorder(root->right);
25     }
26 }
27 void inorder(struct Node *root)
28 {
29     if(root != NULL)
30     {
31         inorder(root->left);
32         printf("%d ", root->data);
33         inorder(root->right);
34     }
35 }
```

C:\Users\Sevanthi-v\Downloads\DS lab 10.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 57, Col 21, Pos 1265 Insert Read/Write default

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Start here x DS lab 9.c x *DSlab 7.c x DS lab8.c x DS lab 10.c x DS lab 11.c x DS lab 12.c x

```
28 {
29     if(root != NULL)
30     {
31         inorder(root->left);
32         printf("%d ", root->data);
33         inorder(root->right);
34     }
35 }
36 void postorder(struct Node *root)
37 {
38     if(root != NULL)
39     {
40         postorder(root->left);
41         postorder(root->right);
42         printf("%d ", root->data);
43     }
44 }
45 int main()
46 {
47     struct Node *root = createNode(1);
48     root->left = createNode(2);
49     root->right = createNode(3);
50     root->left->left = createNode(4);
51     root->left->right = createNode(5);
52     printf("Preorder Traversal:\n");
53     preorder(root);
54     printf("\n\nInorder Traversal:\n");
55     inorder(root);
56     printf("\n\nPostorder Traversal:\n");
57     postorder(root);
58     return 0;
59 }
60
61
```

C:\Users\Sevanthi-v\Downloads\DS lab 10.cC/C++Windows (CR+LF)WINDOWS-1252Line 57, Col 21, Pos 1265InsertRead/Writedefault

```
Preorder Traversal:
1 2 4 5 3

Inorder Traversal:
4 2 5 1 3

Postorder Traversal:
4 5 2 3 1
Process returned 0 (0x0)   execution time : 0.147 s
Press any key to continue.
|
```